

The artifact selected for this Milestone would be the filtering logic utilized within the dashboard application developed in CS340. The dashboard retrieves animal shelter data from a MongoDB database and filters the records based on various rescue training categories. These filters allow the system to display an animal that may be suitable for specific types of search and rescue operations. Currently, this filtering process uses simple database queries that match certain breeds or age ranges. While this approach works for basic filtering, it doesn't provide the most sophisticated method for identifying ideal candidates for training.

I included this algorithm in my eportfolio due to the fact that it demonstrates my skills in designing and implementing data-driven logic as well as applying core programming concepts like data structures and sorting. Some of the key components that showcase my abilities would be the following: the weighted scoring system, data structure usage, and the scoring and ranking system. The weighted scoring system assigns points based on attributes like the breed, age, and traits of the animals within the database. A list of dictionaries is efficiently processed and computed scores are added to each entry. Python's built-in sorted function is utilized with lambda expressions in order to rank animals based on suitability. This artifact was improved from the previous version through the introduction of a more flexible and category-specific scoring system. The inclusion of multiple attributes makes for a better evaluation of the animals and the scoring being integrated allows for ranked outputs to be shown on the dashboard.

I believe that the enhancements made for this milestone meet a couple of different course outcomes including the following:

The enhancement will align with the following course outcomes:

Design and evaluate computing solutions that solve a given problem using algorithmic principles and computer science practices and standards appropriate to its solution, while managing the trade-offs involved in design choices. Demonstrate an ability to use well-founded and innovative techniques, skills, and tools in computing practices for the purpose of implementing computer solutions that deliver value and accomplish industry-specific goals. These are demonstrated through the introduction of the scoring system and the incorporation of algorithmic reasoning and data manipulation, among others.

While creating and refining this algorithm, I learned how to go about designing a weighted scoring system which balances multiple factors. I also learned how to structure data as lists of dictionaries to improve processing. The challenges that I experienced throughout this project would be the process of determining the appropriate weight for each attribute as well as debugging the integration with Mongo. Overall I believe that this artifact strengthened my skills when it comes to working with and designing algorithms.